

Bayesian Method Simulation: Review Results from CSV

Simulation design

Table 1: Simulation settings used in the run.

Setting	Truth	Reader missingness	Z missingness
S1	Correctly specified	10%	0%
S2	Correctly specified	30%	20%
S3	Mildly misspecified	10%	0%
S4	Mildly misspecified	30%	20%

Results

Table 2: Average performance under correctly specified truth.

Setting	Method	AUC	Brier	X cor	X RMSE	C_X MAE
S1	Averaged-reader logistic	0.695	0.211	NaN	NaN	NaN
S1	Bayesian joint model	0.914	0.142	0.584	0.810	0.139
S1	Reader-1 logistic	0.686	0.212	NaN	NaN	NaN
S2	Averaged-reader logistic	0.668	0.220	NaN	NaN	NaN
S2	Bayesian joint model	0.943	0.137	0.563	0.827	0.155
S2	Reader-1 logistic	0.665	0.220	NaN	NaN	NaN

Table 3: Average performance under mildly misspecified truth.

Setting	Method	AUC	Brier	X cor	X RMSE	C_X MAE
S3	Averaged-reader logistic	0.666	0.228	NaN	NaN	NaN
S3	Bayesian joint model	0.908	0.159	0.624	0.773	0.126
S3	Reader-1 logistic	0.664	0.228	NaN	NaN	NaN
S4	Averaged-reader logistic	0.656	0.230	NaN	NaN	NaN
S4	Bayesian joint model	0.938	0.150	0.559	0.831	0.137
S4	Reader-1 logistic	0.651	0.232	NaN	NaN	NaN

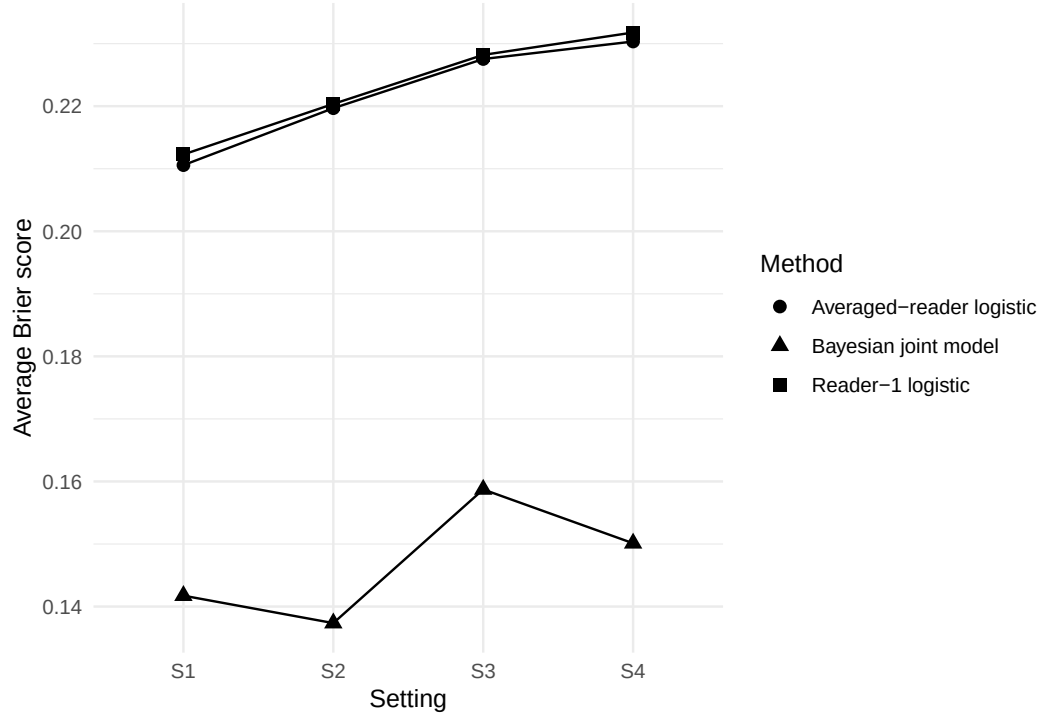
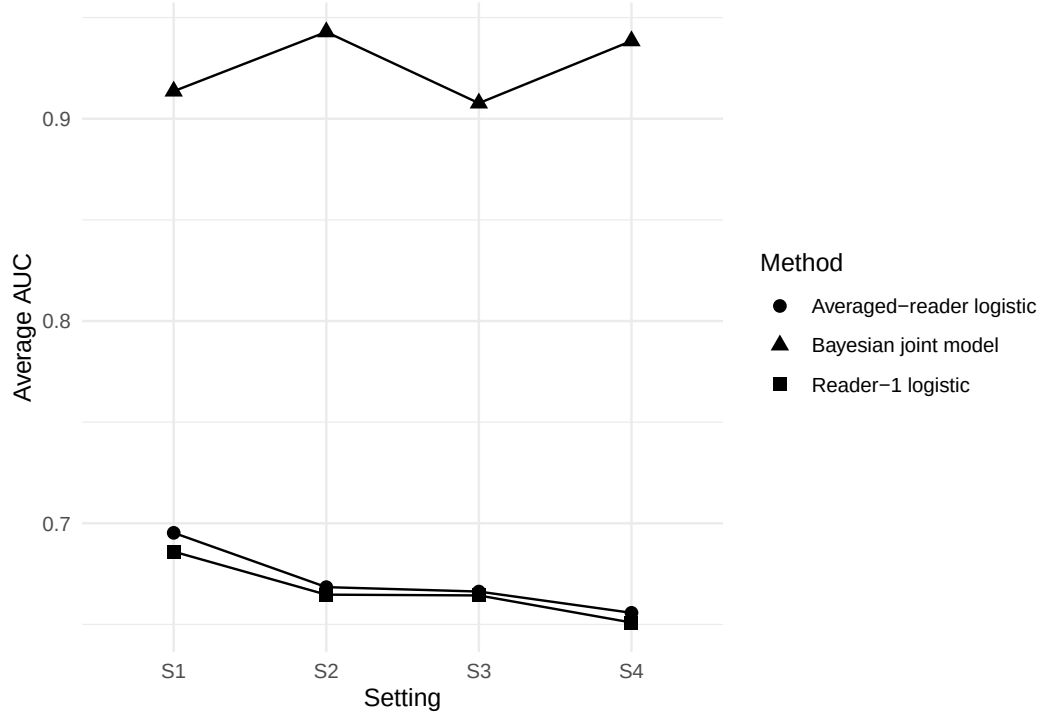


Table 4: Convergence summary for key model parameters.

Setting	Mean key R-hat	Max key R-hat	Mean bulk ESS	Min bulk ESS
S1	1.008	1.054	1328.927	68.167

S2	1.007	1.045	976.081	72.060
S3	1.009	1.073	1383.887	64.850
S4	1.009	1.064	1072.614	90.919
